

Traffic Prediction and Analysis System - Execution Steps

Prerequisites

1. Node.js: Ensure Node.js (v18 or higher) is installed.
2. Python: Ensure Python (v3.8 or higher) is installed.
3. MongoDB: Ensure MongoDB is running or you have a valid MongoDB connection URI.

1. Backend Setup (Node.js)

1. Open a terminal and navigate to the backend directory: ``cd road-traffic-backend-main``
2. Install the required Node.js dependencies: ``npm install``
3. Set up the environment variables:
 - Create a `.env`` file in the ``road-traffic-backend-main`` directory.
 - Add your connection string and variables: ``PORT=4000`, `MONGODB_URI=your_uri``
4. Start the Node.js backend server: ``npm start``
The backend server will run on `http://localhost:4000`.

2. Pothole Detection Model Setup (Python/Flask)

1. Open a new terminal and navigate to the pothole model directory: ``cd road-traffic-backend-main/pothole-model``
2. Install the required Python dependencies: ``pip install flask flask-cors tensorflow pillow numpy``
3. Start the Flask server: ``python app.py``
The pothole detection model will be accessible at `http://localhost:3003`.

3. Frontend Setup (React/Vite)

1. Open a new terminal and navigate to the frontend directory: ``cd road-traffic-frontend-main``
2. Install the required Node.js dependencies: ``npm install``
3. Start the Vite development server: ``npm run dev``
4. Open your web browser and navigate to the URL provided by Vite (typically `http://localhost:5173`).

Summary of Running Services

- Frontend: `http://localhost:5173`
- Main Backend API: `http://localhost:4000`
- Pothole Detection API: `http://localhost:3003`

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You can now use the application from the frontend interface!